

NFL Point Spreads - Solutions

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Solutions to 20 questions from the NFL Point Spreads and Game Scores module.

```
library(Hmisc)
```

Get the data set.

```
NFLPoints<-read.csv("NFLPoints.csv")
```

1. What proportion of the time does the favored team actually win the game outright?

Procedure: CI for a proportion (one sample)

```
NumFavWins<-sum(NFLPoints$FavWin)
n=length(NFLPoints$FavWin)
binconf(NumFavWins,n,method="asymptotic")
```

PointEst	Lower	Upper
0.6801471	0.6409525	0.7193416

We are 95% sure that the proportion of times the favored team wins the game is between 0.641 and 0.719.

2. Is the chance the favorite covers the spread discernibly different from 0.50?

Procedure: Test for proportion (one sample)

$H_0 : p = 0.5$ vs. $H_a : p \neq 0.5$ where p is the proportion of games the favored team covers the spread

```
NumFavCover<-sum(NFLPoints$FavCover)
n=length(NFLPoints$FavWin)
prop.test(NumFavCover,n, correct=FALSE)
```

1-sample proportions test without continuity correction

```
data: NumFavCover out of n, null probability 0.5
X-squared = 3.2426, df = 1, p-value = 0.07174
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.4965875 0.5800770
sample estimates:
      p
0.5386029
```